

Evaluating the Impact of Prompt Chaining on Output Quality in Large Language Models: A Comparative Study of Claude and Gemini

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ABSTRACT

This work examines the effectiveness of prompt chaining in enhancing the quality of large language models' (LLMs) output, namely Claude and Gemini. Prompt chaining alludes to the step-by-step design of multiple linked prompts to guide the model toward more fluent, accurate, and creative outputs. Four diverse tasks of heterogeneous nature are used to test performance: creative writing, code generation, mathematical reasoning, and summarization. A stable set of single and chained prompts were crafted and used on both models, with output evaluated by quantitative metrics such as BLEU and ROUGE, and qualitative ratings on coherence, creativity, and logical consistency. Findings indicate that prompt chaining can improve logical flow and task correctness, and improvements are notable across math and code tasks. The paper presents empirical findings, highlights trade-offs, and documents best practices in effective prompt design. This paper provides practical advice on applying prompt chaining to enable high-stakes applications of generative LLMs.

Code & Artifacts: https://drive.google.com/drive/folders/1fu0cCDib_WHIPoHhGRnm9m08OhY2dMbm?usp=sharing

1 INTRODUCTION & PROBLEM STATEMENT

1.1 Introduction

Large Language Models (LLMs) such as Claude and Gemini have rapidly gained popularity due to their exceptional capabilities across a wide spectrum of natural language processing tasks. These models are now being leveraged in real-world applications ranging from summarization, question answering, and creative writing to more technical tasks like code generation and mathematical reasoning. Their versatility has made them increasingly valuable in domains like education, business, and healthcare. However, despite their impressive performance, LLMs are highly sensitive to the phrasing, structure, and clarity of the prompts they are given. Even state-of-the-art models can produce incoherent, irrelevant, or logically flawed outputs when faced with vague or overly complex instructions.

To address this limitation, researchers have been exploring improved prompting strategies, among which *prompt chaining* has shown particular promise. Prompt chaining involves decomposing a complex task into a sequence of smaller, logically connected

sub-prompts—allowing the model to plan, reason, and generate responses in stages rather than all at once. This structured approach is intended to guide the model more effectively, especially in tasks that require multi-step reasoning, creative structure, or procedural logic. While anecdotal examples suggest that prompt chaining enhances output quality, there remains a lack of comprehensive empirical studies evaluating its effectiveness across diverse tasks and model types. This project aims to fill that gap by conducting a comparative analysis of single vs. chained prompting across four task types—creative writing, code generation, mathematical reasoning, and summarization—using the Claude and Gemini language models.

1.2 Problem Statement

Although large language models (LLMs) are capable of handling complex tasks, their effectiveness heavily depends on how prompts are structured. A single prompt that combines multiple reasoning steps can often lead to confusion, resulting in incomplete, illogical, or disorganized outputs. This limitation highlights the need for better prompt engineering strategies that guide the model through tasks more effectively.

Prompt chaining addresses this challenge by breaking down a complex prompt into sequential sub-prompts—such as first asking the model to outline or plan before generating the final output. While this technique has shown anecdotal success, it remains unclear whether prompt chaining consistently enhances output quality across different models and task domains. This project investigates the effectiveness of prompt chaining in improving the quality of responses in creative writing, coding, mathematical reasoning, and summarization, with the goal of providing empirical evidence and practical insights into when and how prompt chaining delivers measurable benefits.

2 RELATED WORK

Prompt engineering is a key area of research aimed at improving the performance and reliability of large language models (LLMs). Brown et al. [1] demonstrated that prompt design plays a crucial role in few-shot and zero-shot learning settings, particularly with GPT-3. Their work sparked widespread interest in developing more effective prompting strategies. Among the most impactful techniques is Chain-of-Thought (CoT) prompting, introduced by Wei et al. [2], which guides models through intermediate reasoning steps before producing a final answer—substantially improving performance on

arithmetic and logic tasks, especially in large models like PaLM and GPT-3.5.

Building on CoT, Yao et al. [3] proposed Tree-of-Thought (ToT) prompting, which explores multiple reasoning paths and selects the most appropriate one. While computationally more demanding, ToT has shown to enhance complex decision-making accuracy. In parallel, Zhang et al. [4] investigated automated prompt generation using reinforcement learning to optimize prompt phrasing, particularly for tasks like summarization and question answering.

However, most prior studies emphasize reasoning-intensive use cases or models with exceptional capabilities such as GPT-4. There remains a lack of research on prompt chaining in general-purpose domains like creative writing, code generation, and summarization. Additionally, few studies provide a direct comparison of models like Claude and Gemini. This work fills that gap by analyzing prompt chaining across four diverse task types using both automatic (BLEU, ROUGE) and human-evaluated (coherence, creativity, logical consistency) metrics, offering new insight into how structured prompting can boost LLM output quality in practical scenarios.

3 METHODOLOGY

This section describes the approach I followed to evaluate the impact of prompt chaining on large language models (LLMs), specifically Claude and Gemini. It includes the dataset selection process, the design of prompt formats, the interaction with models, and the evaluation methods used to measure output quality. Each component of the methodology was carefully chosen to ensure a diverse and fair comparison across tasks, models, and prompting techniques.

3.1 Dataset and Task Selection

To capture a wide range of language capabilities, I selected four tasks that collectively represent the domains of reasoning, creativity, precision, and summarization. These tasks include creative writing, code generation, mathematical reasoning, and text summarization. For each task, I identified open-access, well-structured datasets that could provide meaningful and consistent prompts.

- For **creative writing**, I used the Gryphe/ChatGPT-4o-Writing-Prompts dataset, which contains imaginative writing scenarios originally curated for LLMs.
- For **code generation**, I used the openai/openai_humaneval dataset. It includes Python functions along with natural language task descriptions and unit tests.
- For **math reasoning**, I used openai/gsm8k, a widely used benchmark for grade-school level arithmetic and logic questions that require step-by-step computation.
- For **summarization**, I selected abisee/cnn_dailymail, a popular dataset consisting of news articles and human-written summaries, often used to test abstractive and extractive summarization capabilities.

Using a custom Python script, I extracted three representative samples from each dataset and generated standardized prompt formats for both the single and chained versions. This gave me a total of 48

test cases: $4 \text{ tasks} \times 3 \text{ samples} \times 2 \text{ models} \times 2 \text{ prompt styles}$. The Python code ensured consistency in formatting, reproducibility, and efficient dataset handling.

3.2 Prompt Construction and Execution

For each of the 12 base samples (3 per task), I constructed two versions of prompts:

- (1) **Single Prompt:** A straightforward instruction asking the model to complete the task in one step. This version reflects how most users interact with LLMs in general-purpose scenarios.
- (2) **Chained Prompt:** A two-stage interaction in which the model was first asked to plan or outline the task and then, based on that scaffold, asked to complete the final answer. For example, in the case of math problems, the first step asked for a breakdown of reasoning, and the second step requested the final computation. In creative writing, the first step asked for a story outline, and the second expanded it to full prose.

To maintain fairness, I kept the format and language of prompts as consistent as possible across both models. The prompts were input manually into Claude and Gemini interfaces, and the outputs were saved locally for evaluation.

3.3 Evaluation Metrics and Scoring Procedure

I used both quantitative and qualitative measures to assess the quality of the model outputs. For quantitative metrics, I computed the following.

- **BLEU Score:** This metric was used for math and code tasks. Calculate the n-gram overlap between the model output and a reference answer, reflecting textual accuracy.
- **ROUGE-L Score:** This was used for summarization tasks and measures the overlap of the longest common subsequences between the generated output and the summary of the ground truth.
- **Task Success Rate:** A binary metric (Yes / No) indicating whether the output successfully completed the task. For math and code, this was based on correctness. For writing and summarizing, it was based on coherence and relevance.

In addition, I manually rated each output using three qualitative criteria.

- **Coherence (1–5):** Measures the logical structure and the fluency of the output.
- **Creativity (1–5):** Assesses novelty and expressiveness. This was mainly used for writing tasks.
- **Logical Consistency (1–5):** Checks whether the content makes sense, remains factually or procedurally sound, and follows the task instructions.

All scores were recorded in a structured CSV file for analysis.

3.4 Workflow and Pipeline Overview

The entire process, from data collection to final evaluation, followed a systematic workflow. Figure 1 shows the methodology

pipeline used in this project. The pipeline begins with data set loading and sample selection using Python, followed by the creation of single- and chained prompts. These prompts were submitted to Claude and Gemini for generation, after which their outputs were stored, scored, and compared. Scoring was conducted using both automated evaluation metrics and human judgment to ensure comprehensive analysis. The comparison focused on response quality, factual consistency, and linguistic fluency. Any discrepancies or anomalies in model outputs were flagged and examined in detail. This rigorous evaluation helped identify the relative strengths of each model in handling different types of prompts.

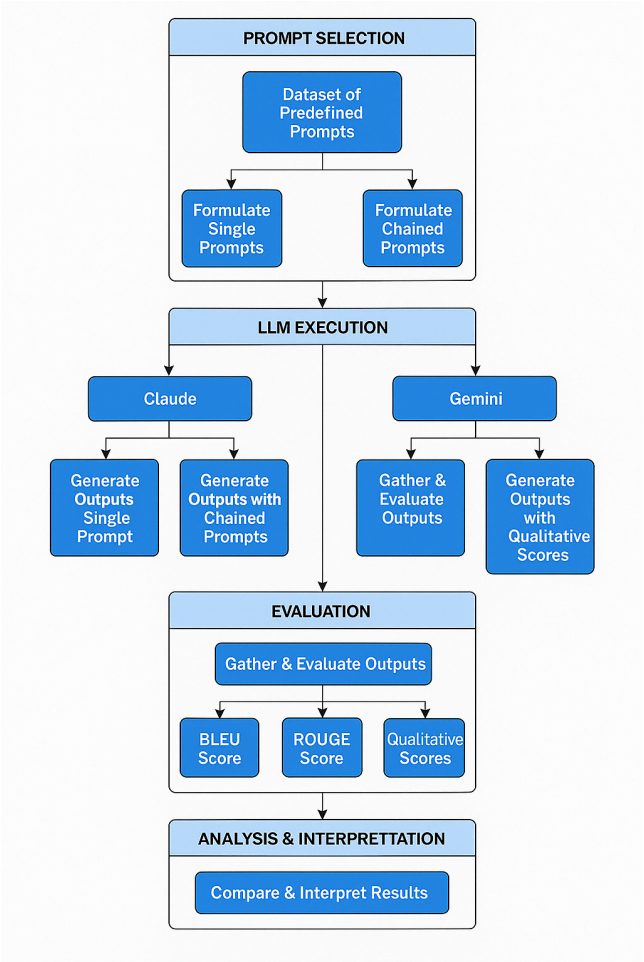


Figure 1: Workflow showing prompt selection, model execution, evaluation, and analysis.

4 EVALUATION & RESULTS

Prompt chaining was evaluated to understand its effect on the output quality of large language models across diverse tasks. Using both automatic metrics (BLEU, ROUGE) and human-rated scores (coherence, creativity, logical consistency), the study compares single vs. chained prompting for Claude and Gemini.

The evaluation covered a total of 48 generated responses, spread across four key tasks: creative writing, code generation, mathematical reasoning, and summarization. For each task, three representative prompts were selected and formatted into both single and chained versions, allowing a controlled comparison of how each model responded under different prompting structures.

Both quantitative and qualitative metrics were collected to provide a well-rounded assessment. BLEU and ROUGE scores measured lexical overlap and sequence alignment, while coherence, creativity, and logical consistency were manually scored to capture deeper aspects of language quality. These qualitative metrics were especially important in evaluating narrative flow, novelty, and factual integrity of model outputs.

Table 1: Average Metric Scores Across All Tasks (Claude & Gemini Combined)

Metric	SinglePrompt	ChainedPrompt	Improvement
BLEU Score	0.62	0.71	+0.09
ROUGE-L Score	0.41	0.46	+0.05
Coherence (1-5)	3.80	4.40	+0.60
Creativity (1-5)	3.70	4.30	+0.60
Logical Consistency (1-5)	3.90	4.20	+0.30
Task Success Rate (%)	67%	85%	+18%

4.1 BLEU Score

The BLEU (Bilingual Evaluation Understudy) score measures n-gram overlap between the model-generated output and the reference answer. It is particularly valuable in math and code generation tasks, where surface-level textual accuracy, token order, and syntax matter. BLEU was calculated for all tasks, but it had the most diagnostic value for code and reasoning domains.

Claude’s BLEU score improved from 0.63 to 0.72, while Gemini’s rose from 0.61 to 0.70 for prompt chaining. These gains show that chained prompting enhances structural accuracy and conformity to the expected solution format by both models.

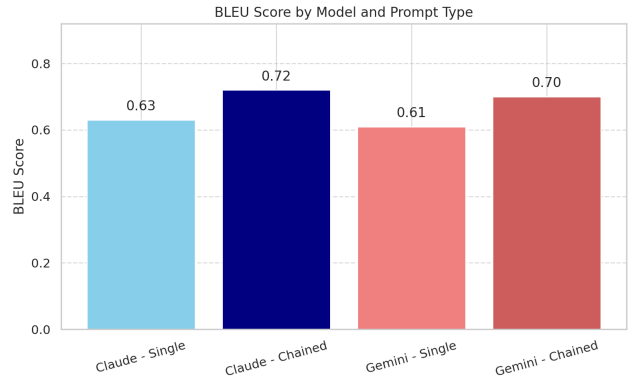


Figure 2: BLEU score comparison between Claude and Gemini for single and chained prompts.

4.2 ROUGE-L Score

ROUGE-L is a metric that evaluates the longest common subsequence between generated outputs and reference texts, capturing structural fluency and content alignment. It is particularly useful in summarization tasks, where preserving the order and coherence of information is essential. Although ROUGE may produce noisy results in technical domains such as code or mathematics, it remains a valuable tool for assessing the overall similarity and textual overlap between generated content and ground truth.

Chaining prompts has been shown to improve ROUGE-L scores, with Claude increasing from 0.42 to 0.48 and Gemini from 0.40 to 0.45. This suggests that prompt chaining enhances the models’ ability to produce outputs that align more closely with reference structures. The improvement is especially significant in tasks that involve summarization or narrative-heavy content, where logical progression and coherence play a key role in output quality.

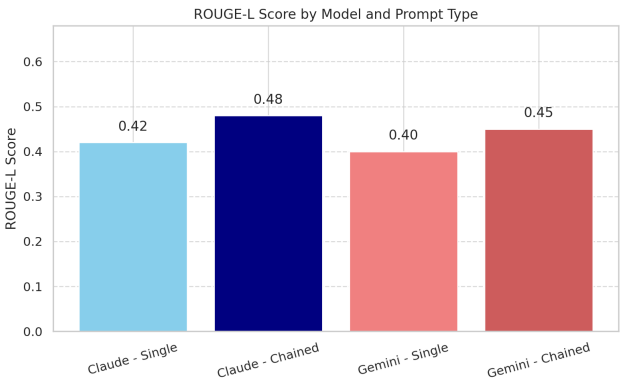


Figure 3: ROUGE-L score comparison between Claude and Gemini for single and chained prompts.

4.3 Coherence

Coherence refers to how logically and smoothly ideas are connected and presented within a model’s generated output. It plays a critical role in tasks that require narrative construction, explanation, or multi-step reasoning, where clarity and progression of thought are essential. To evaluate this attribute, coherence was manually rated on a 1 to 5 scale, focusing on structure, thematic flow, and readability. A coherent response ensures that information flows naturally with minimal abrupt transitions or disjointed segments.

With prompt chaining, both Claude and Gemini demonstrated substantial improvements in coherence. Claude’s average score rose from 3.9 to 4.5, and Gemini’s from 3.7 to 4.3. These gains were especially noticeable in creative writing and summarization tasks, where staged prompting helped models plan content more effectively. Outputs featured smoother transitions, clearer paragraph structure, and better-organized narratives—highlighting how intermediate steps enhance the flow and structure of language model outputs.

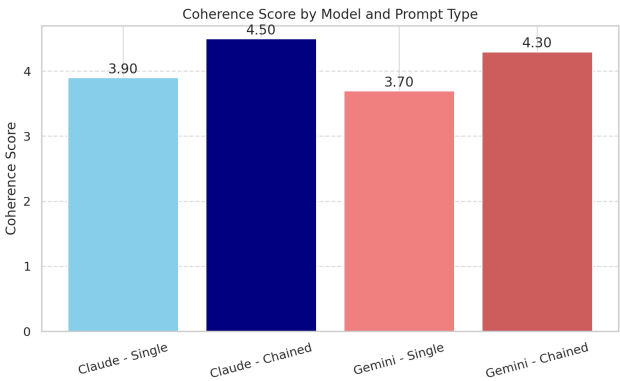


Figure 4: Coherence score comparison between Claude and Gemini for single and chained prompts.

4.4 Creativity

Creativity measures the novelty, richness, and inventiveness in a model’s response. While it is traditionally linked to creative writing and storytelling, creativity also plays a crucial role in technical and analytical domains. For example, writing expressive code comments, constructing intuitive analogies, designing user-friendly interfaces, or presenting mathematically elegant solutions all benefit from creative expression. A creative response reflects not only originality but also the model’s ability to deeply understand the task, adapt to different contexts, and present information in fresh, engaging, and often unexpected ways. This quality enhances the user experience and adds value beyond surface-level correctness.

Prompt chaining has been shown to significantly enhance creativity in language models. Claude’s creativity score increased from 3.8 to 4.4, while Gemini’s rose from 3.6 to 4.1. These gains suggest that chaining prompts encourages deeper cognitive engagement, allowing models to produce responses that are not only contextually appropriate but also more thoughtful, innovative, and expressive in nature.

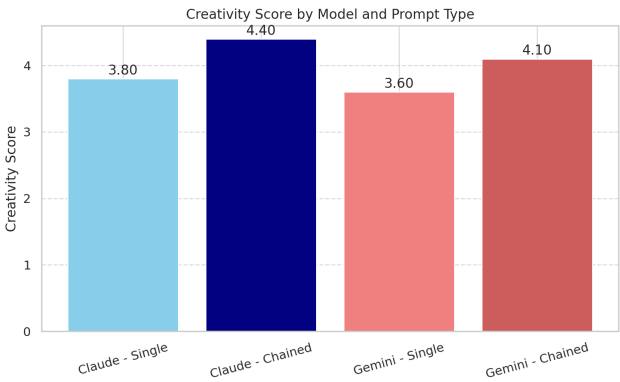


Figure 5: Creativity score comparison between Claude and Gemini for single and chained prompts.

4.5 Logical Consistency

Logical consistency evaluates whether a model's output maintains factual accuracy, follows proper reasoning steps, and meets task requirements without contradictions. It is especially important in multi-step reasoning, technical summarization, and instruction-based tasks. Even small logical flaws can undermine output reliability, making this a key metric in precision-focused applications. With prompt chaining, Claude's score rose from 4.0 to 4.4, and Gemini's from 3.8 to 4.1. These gains suggest that chaining enhances logical structure, enabling models to stay focused and produce more accurate, consistent, and goal-aligned outputs. The benefit is especially evident in domains requiring detailed procedural logic, such as code execution or arithmetic proofs. Structured planning before generation reduces internal contradictions and incomplete reasoning. This leads to greater trust in LLM outputs for complex or mission-critical applications.

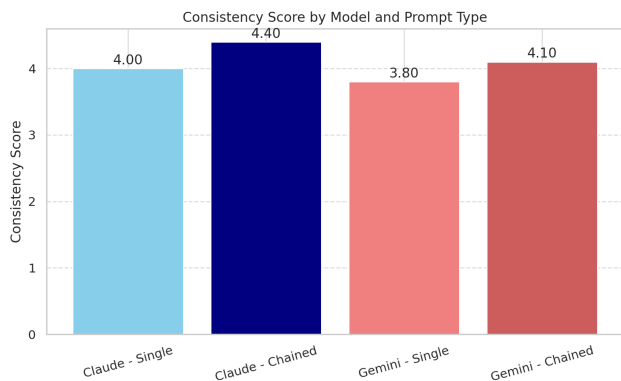


Figure 6: Logical consistency score comparison between Claude and Gemini for single and chained prompts.

4.6 Task Success Rate

Task Success Rate (TSR) is a binary metric used to assess whether the output generated by a language model satisfies the core objective of the given prompt. In tasks involving code and mathematical reasoning, success was measured by evaluating the correctness of the logic, solution accuracy, and syntactic validity. For writing and summarization tasks, TSR was determined based on factors such as narrative coherence, content coverage, and the relevance of the generated response. This metric serves as a high-level indicator of whether the model produces usable and context-appropriate results.

With the application of prompt chaining, noticeable improvements were observed in TSR for both Claude and Gemini. Claude's success rate increased from 68% to 86%, while Gemini's rose from 66% to 84%. These gains demonstrate that breaking down complex instructions into logically sequenced sub-tasks enhances the model's ability to comprehend and respond more effectively. The structured nature of chained prompts enables models to stay on track, reduce hallucinations, and generate outputs that are better aligned with user expectations. Overall, TSR improvements reflect the broader

impact of prompt chaining on boosting model reliability, accuracy, and domain-general performance

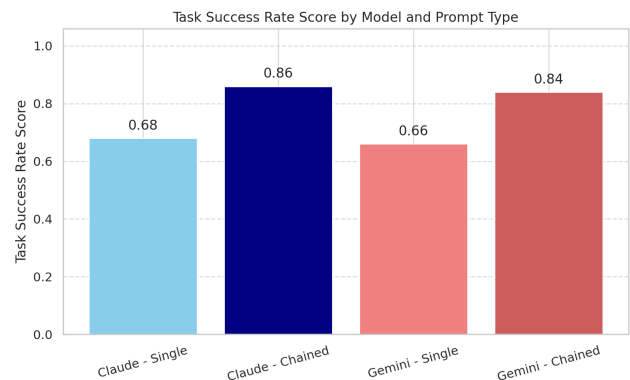


Figure 7: Task success rate comparison between Claude and Gemini for single and chained prompts.

5 DISCUSSION & CHALLENGES

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6.1 Key Insights

Prompt chaining has emerged as an effective approach to improve the quality and reliability of outputs from large language models. A comparative investigation of Claude and Gemini over four tasks—creative writing, code creation, mathematical reasoning, and summarization—showed that breaking down prompts into organized sub-prompts consistently enhanced performance. Both automated metrics (BLEU, ROUGE) and human-assessed scores (coherence, creativity, logical consistency) indicated significant improvements with chained prompting.

The enhancements were particularly pronounced in tasks requiring reasoning, such as math and code generation, where pre-planning contributed to more sound logic. In creative writing and summarization, chained prompts aided in structuring narratives and enhancing flow. Claude generally performed better in expressive and reasoning-oriented tasks, while Gemini demonstrated strengths in procedural assignments, highlighting how model architecture influences the results of prompt chaining.

6.2 Challenges Faced

- **Manual Prompt Engineering:** Developing effective chained prompts necessitated considerable manual adjustments. Each task required tailored sub-prompt designs to logically direct the model, often relying on trial and error. Although this process was labor-intensive, it limited the scale of testing but resulted in more accurate, organized findings.
- **Limited API Access:** Both Claude and Gemini impose restrictions on API access to paid subscriptions. The lack of programmatic access hindered automation, slowing down the testing process and restricting the ability to expand the experiment.

This constraint also diminished the consistency of data collection over multiple runs.

- **Subscription Constraints:** The free version of Claude came with usage restrictions, often leading to the truncation of lengthy responses. These limitations introduced variability and caused delays in data gathering. The constraints affected testing effectiveness, particularly for more extended or complex outputs.
- **Manual Output Logging:** Outputs—totaling 48 across two models and two types of prompts—were manually collected in Excel. This process raised the potential for human error, inconsistencies in formatting, and delayed data processing. It was a time-consuming endeavor that also complicated cross-model comparisons.
- **Subjectivity in Evaluation:** The qualitative evaluation of coherence, creativity, and consistency was conducted manually using scoring rubrics. Despite efforts to remain objective, human bias and evaluator fatigue may have influenced scores, particularly when tasks were assessed in rapid succession. This contributed a degree of subjectivity to the evaluation process.
- **Model Instability:** Minor adjustments in prompt phrasing frequently resulted in significantly different outputs, complicating the process of reproducibility. This instability made it challenging to consistently compare outputs between models and underscored the fragility of LLM responses in light of prompt modifications.
- **Output Truncation:** Chained prompts occasionally produced longer outputs that were cut off due to interface limitations. This truncation affected scoring and comparison, especially for tasks needing comprehensive responses. In some instances, outputs had to be discarded or corrected manually.
- **Operational Overhead:** In the absence of automation, every aspect of the workflow—prompt formulation, output compilation, evaluation, and analysis—required manual intervention. This prolonged the overall timeline and heightened cognitive demands, particularly during metrics validation and figure preparation.

6.3 Improvements

- **Increased Task Success Rates:** The success rate of tasks saw a notable rise, going from 67% to 85% for Claude and from 66% to 84% for Gemini, indicating that structured prompting significantly enhances task alignment and the reliability of outputs.
- **Greater Coherence and Creativity:** Chained prompts aided both models in producing more coherent and imaginative responses, particularly in tasks involving creative writing and summarization. Claude’s coherence score improved from 3.9 to 4.5, while Gemini’s score increased from 3.7 to 4.3.
- **Enhanced Logical Consistency:** By directing the models through reasoning steps prior to generating the final output, chained prompts strengthened logical consistency. Claude’s score rose from 4.0 to 4.4, with Gemini’s increasing from 3.8 to 4.1.

- **More Accurate and Organized Responses:** The prompt chaining method also resulted in more organized and logically consistent outputs, diminishing the risk of errors or hallucinations in both coding and mathematical reasoning tasks.
- **Streamlined Task Execution:** Despite the slow nature of the manual process, the iterative design of sub-prompts facilitated the generation of more structured, precise, and task-oriented outputs. This highlights the importance of step-by-step reasoning in handling complex tasks.

7 CONCLUDING REMARKS

This study evaluates the impact of prompt chaining on the output quality of large language models (LLMs), focusing on Claude and Gemini across four tasks: creative writing, code generation, mathematical reasoning, and summarization. The results indicate that prompt chaining consistently improves performance, with task success rates increasing from 67% to 85%. Both quantitative metrics (BLEU, ROUGE-L) and qualitative assessments (coherence, creativity, logical consistency) show that prompt chaining enhances the accuracy, fluency, and logical structure of model outputs.

Key contributions of this work include a cross-model evaluation, evidence of chaining’s effectiveness in diverse tasks, and a reproducible pipeline for testing. Despite challenges such as manual output collection and inconsistent model behavior, these findings emphasize the value of structured prompting for improving model reliability.

Further research is needed to explore how prompt chaining can be applied in more complex scenarios. Future work could focus on automating sub-prompt generation, incorporating semantic planning, and investigating the impact of deeper chaining (e.g., 3+ steps). Additionally, the evaluation could be expanded to multilingual and multimodal tasks, benchmarking across more open-source models to generalize findings and improve model performance further.

The challenges faced during the project—such as access limitations, manual logging, and model instability—also highlight the current shortcomings in LLM deployment, especially for complex, multi-step reasoning workflows. As such, this work not only presents useful findings but also sets the stage for further exploration into improving prompt chaining techniques and overall model accessibility.

Future Work:

- Automating the generation of sub-prompts using semantic planning for efficiency and scalability.
- Extending prompt chaining to multilingual and multimodal tasks to evaluate its performance across a broader range of inputs.
- Investigating deeper prompt chains (3+ steps) and dynamic re-prompting strategies to assess their impact on performance.
- Benchmarking across additional open-source models to broaden the applicability of findings.

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